Shaik. Jabbar Basha - 192325059.

Q1) A coin is tossed 10 times with outcomes:

H H T H H H T H H T.

Estimate the probability of getting Heads (P) using MLE.

Soultion:-

number of Heads = 7

number of Tails = 3

total tosses = 10

for a Bernouli distribution,

$$\boxed{\hat{P} = \frac{\text{number of Heads}}{\text{Total Tosess}}}$$

$$\hat{P} = \frac{7}{10} = 0.7$$

Q2) out of 50 students, 42 passed. Find the MLE of the probability that a student passes.

Sol:-

passed = 42

total students = 50

using MLE

$$\hat{P} = \frac{\text{Number of Heads}}{\text{Tatal Tosses}}$$

$$\Rightarrow \frac{42}{50} = 0.84.$$

Wnew = 2 (- (0.01 × 4)

Wnew = 2 - 0.04

= 1.96

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Q3) A factory produces light bulbs.

Sample size = 100 bulbs

Defective bulbs = 8

find the MLE of the defective probability.

Sol:-

Defective bulbs = 8

Total bulbs = 100

$$\hat{p} = \frac{8}{100} = 0.08$$

$$\boxed{\hat{p} = 0.08}$$

Interpretation:-

The estimated probability that a bulb is deffective is 8%

Q4) A die is rolled 60 times.

Number 6 appears 18 times.

Estimate the probability of rolling a 6.

Solution:-

Number of sixes = 18.

Total rolls = 60

$$\hat{p} = \frac{18}{60} = 0.30$$

final answer.

$$\hat{p} = 0.30$$

Interpretation:-

The Estimated probability of rolling a 6 is 30%

Q5) Neural Network prediction and Gradient Descent

given :-

input $x = 4$

weight $w = 2$

Actual output: $y_{actual} = 10$

prediction formula

$$y = w \times x$$

Gradient = 4

a) Predicted output :-

$$y = 2 \times 4 = 8$$

b) Error :-

Error = Actual - predicted

$= 10 - 8 = 2$

c) updated weight :-

Gradient Descent formula

$$w_{new} = w - \eta \times \text{gradient}$$

where
* learnin rate $(\eta) = 0.01$
* Gradient = 4

$w_{new} = 2 \; (- (0.01 \times 4))$

$w_{new} = 2 - 0.04$

$= 1.96$

Interpretation :-

The weight is reduced from 2.00 to 1.96, helping the model minimize prediction error.